

COMPUTER ORGANIZATION AND ARCHITECTURE (CSE23CT201)

COURSE OUTCOME	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Study of the basic structure and operation of a digital computer system	H	M	M									
Designing and constructing a computer processor including memory.	H	H	M									
Learning the concept of pipelining for consistent execution of instructions with minimum hazards.	H	M	M						L			
Ability to understand the concept of memory organization.	H	M	M						L			
Understanding the various ways of communicating with i/o devices and standard i/o interfaces	H	M	M						L			

Second Year - Semester III

C. No.	Course Code	Course Category	Course Title	L	T	P/RP	Total Credits	Total Hours/ Marks
27.	CSE23CT201	CT	Computer Organization and Architecture	3	-	-	3	45 / 100

Learning Outcomes

At the end of the course, the student should be able to

CO1: Study of the basic structure and operation of a digital computer system

CO2: Designing and constructing a computer processor including memory.

CO3: Learning the concept of Pipelining for consistent execution of instructions with minimum hazards.

CO4: Ability to understand the concept of memory organization.

CO5: Understanding the various ways of communicating with I/O devices and standard I/O interfaces

Unit	Topics	Specific Learning Objectives
1	Basic Structure of Computers(9 hours) Functional Units, Basic Operational Concepts, Bus Structures, Memory Location and Addresses, Memory operations, Instructions and Instructions Sequencing.	To identify the purpose and role of each functional unit in the overall operation of a computer.

2	Processing Units(9 hours) Register Transfers, Performing of arithmetic or logic operations. Fetching a word from memory, Storing a word in memory. Execution of Complete Instructions. Multiple Bus Organization, Micro programmed Control	To understand register transfers in the context of computer architecture.
3	Pipelining(9 hours) Role of Cache memory, Pipeline performance. Data Hazards. Instructions Hazards, Influence on Instruction sets, Data path and Control Considerations.	To be familiar of the role the cache memory system performance.
4	Memory Organizations(9 hours) Memory Hierarchy, Main memory: RAM and ROM chips. Auxiliary Memory, Associative memory, Cache memory Virtual Memory, Memory Management Hardware.	To analyze the characteristics of memory hierarchy, main memory, Auxiliary memory, with speed, and capacity.
5	Input and Output Organizations(9 hours) Accessing Input/ Output Devices: Interrupts, Interrupt Hardware, Enabling and disabling interrupts, Handling multiple devices, Controlling device requests, Exceptions, Use of Interrupts in Operating system. Direct Memory Access: Bus Arbitration, Buses: Synchronous and Asynchronous Bus. Standard I/O Interfaces: Peripheral Component Interconnect Bus, SCSI Bus.	To be familiar with enabling and disabling input and output interrupts.

TEXTBOOK REFERENCES:

1.	1. Hamacher, C. Vranesic Z and Zaky S (2002). Computer organization and embedded systems. McGraw Hill Publication Fifth edition.
2.	2. M.Moris Mano Computer System Architecture (PHI publication)

OTHER REFERENCES:

1.	1. Andrew S. Tanenbaum Structured Computer Organization (2016.) Pearson Education India
2.	2. Hennessy, J. L., & Patterson, D. A. (2011). Computer architecture: a quantitative approach. Elsevier.
3.	3. William Stallings, Computer Organization and Architecture -- Designing for Performance, Pearson Education.